



GIRRAWEE HIGH SCHOOL

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

2002

MATHEMATICS

EXTENSION 2

*Time allowed - Three hours
(Plus 5 minutes' reading time)*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are supplied
- Board-approved calculators may be used.
- Each question attempted is to be returned on a *separate* piece of paper clearly marked Question 1, Question 2, etc. Each piece of paper must show your name.
- You may ask for extra pieces of paper if you need them.

QUESTION 1

a) Find $\int \frac{dx}{\sqrt{x^2 + 2x + 5}}$

2

b) (i) Find real numbers a, b and c such that

$$\frac{3x}{(x+1)(x^2+2x+4)} \equiv \frac{a}{x+1} + \frac{bx+c}{x^2+2x+4}$$

2

(ii) Find $\int \frac{3x}{(x+1)(x^2+2x+4)} dx$

2

c) Use integration by parts to find

$$\int_0^1 \tan^{-1} x \, dx$$

3

d) Find $\int_0^{\frac{2\pi}{3}} \frac{1}{5+4\cos x} dx$

3

e) Find $\int \sin^3 x \cos^2 x \, dx$

3

QUESTION 2

(a) Given the two complex numbers $z = 3-4i$ and $w = 4+3i$, find zw and $\frac{1}{w}$ in the form $x+iy$.

2

b) On separate argand diagrams draw a neat sketch of the locus specified by

(i) $z^2 - \bar{z}^2 = 4i$

2

(ii) $\arg(z-2) = \arg z + \frac{\pi}{2}$

2

c) If $z = \sqrt{3} + i$

(i) Find the exact value of $\text{mod } z$ and $\arg z$.

2

(ii) By using De Moivre's theorem write $\frac{1}{z^5}$ in form $x+iy$.

2

d) Let P, Q, R represent the complex numbers z_1, z_2, z_3 respectively.

What geometric properties characterize triangle PQR if $z_2 - z_1 = i(z_3 - z_1)$?

Give reasons for your answer.

3

e) The polynomial $z^3 - 3z^2 + 7z - 5$ has one root equal to $1-2i$.

Factorize this polynomial

2

QUESTION 3

(a) An ellipse has the equation $\frac{x^2}{25} + \frac{y^2}{16} = 1$

(i) Find the eccentricity.

1

(ii) Find coordinates of the foci S, S' and equation of directrices.

2

(iii) Sketch the ellipse showing all the above features and where it crosses the coordinate axes.

1

(iv) If P is a point on the ellipse show that PS + PS' is independent of the position of P.

2

(b) Consider the hyperbola with equation $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

(i) Show that the equation of the tangent at the point P (asec θ , btan θ) has the equation $bx \sec \theta - ay \tan \theta = ab$.

2

(ii) Deduce the equation of the normal at P.

2

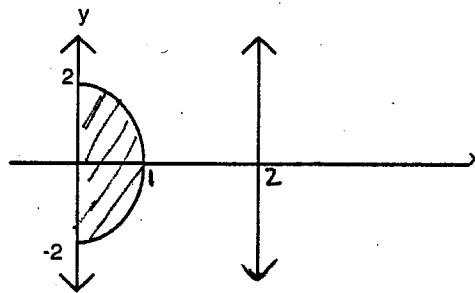
(iii) Find A and B where the tangent and normal respectively cut the y-axis.

2

(iv) Show that AB is the diameter of the circle that passes through the foci of the hyperbola.

3

QUESTION 4



(a) A solid S is formed by rotating the region bounded by the parabola $y^2 = 4(1-x)$ and the y -axis 360° about the line $x = 2$.

(i) By slicing perpendicular to the axis of rotation, find the exact volume of S .

4

(ii) (a) Use the method of cylindrical shells to show that the volume of S is also

given by $\int_0^1 8\pi(2-x)\sqrt{1-x} \, dx$.

2

(b) Confirm your answer to part (i) by calculating this definite integral using the substitution $u = 1-x$.

3

(b) A dome has a circular base of radius 10 metres. Each cross section of the dome perpendicular to the x -axis is a parabola, whose height is the same as the base width.

(i) Why would Simpson's rule give the exact area of the parabolic cross section?

1

(ii) Show that the area of the parabolic cross-section is $\frac{8y^2}{3}$ square metres.

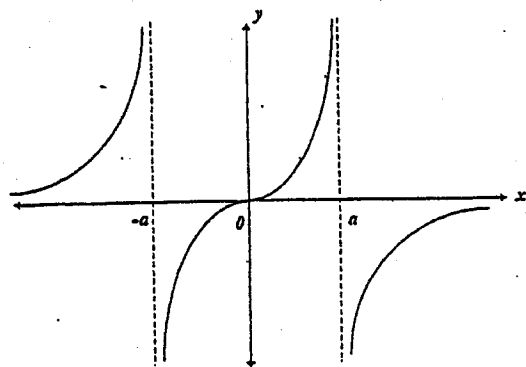
2

(iii) Find the volume of the dome.

3

QUESTION 5

(a) The graph of $y=f(x)$ is shown below



Draw sketches of the following

(i) $y=f(x-a)$

1

(ii) $y=f'(x)$

2

(iii) $y=\frac{1}{f(x)}$

2

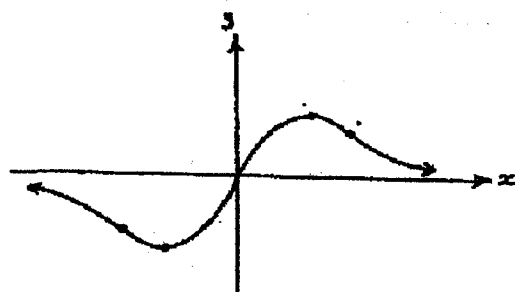
(iv) $y=f(x)^2$

2

(b) Find integers a and b such that $(x+1)^2$ is a factor of $x^3 + 4x^2 + ax + b$

3

(c)



The curve $y=\frac{2x}{1+x^2}$ is sketched in the diagram above

(i) Show that the equation $kx^3 + (k-2)x = 0$ can be written in the form $\frac{2x}{1+x^2} = kx$

2

(ii) Using a graphical approach based on the curve $y=\frac{2x}{1+x^2}$, or otherwise, find

the real values of k for which the equation $kx^3 + (k-2)x = 0$ has exactly 1 solution.

3

QUESTION 6

(a) A particle of mass m is projected vertically upwards under gravity

The air resistance to the motion is $-\frac{1}{100}mgv^2$ where v is the speed of the particle

(i) Show that during the upward motion of the particle, if x is the upward vertical displacement of the particle from its projection point at time t then

$$\ddot{x} = -\frac{1}{100}g(100 + v^2). \quad 2$$

(ii) If the speed of projection is u show that the greatest height (above the point of projection) reached by the particle is

$$\frac{50}{g} \ln \left(\frac{100 + u^2}{100} \right). \quad 4$$

(b) Let ω be a non-real cube root of unity.

(i) Show that $1 + \omega + \omega^2 = 0$. 1

(ii) Hence simplify $(1 + \omega)^2$. 1

(iii) Show that $(1 + \omega)^3 = -1$. 1

(iv) Use part iii) to simplify $(1 + \omega)^{3n}$ and hence show that

$${}^{3n}C_0 - \frac{1}{2}({}^{3n}C_1 + {}^{3n}C_2) + {}^{3n}C_3 - \frac{1}{2}({}^{3n}C_4 + {}^{3n}C_5) + {}^{3n}C_6 - \dots \dots {}^{3n}C_{3n} = (-1)^n. \quad 3$$

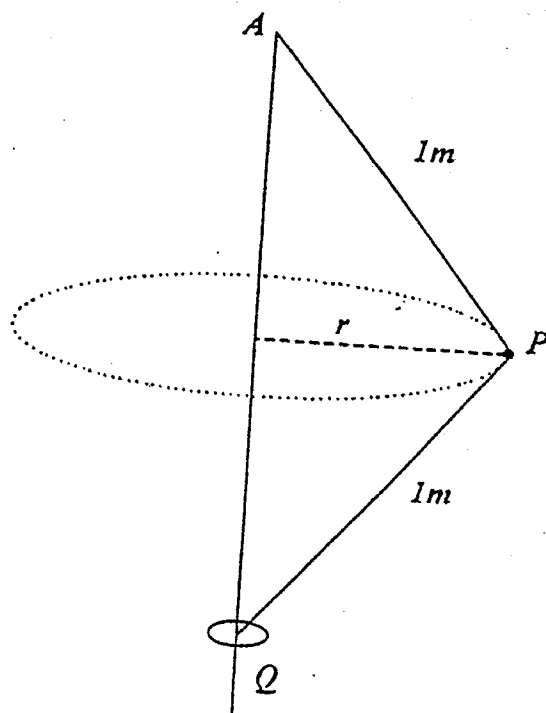
(HINT: You may assume $\operatorname{Re}(\omega) = -\frac{1}{2}$ and that $\operatorname{Re}(\omega^2) = -\frac{1}{2}$)

(c) (i) Show that for $a > 0$ and $n \neq 0$, $\log_{a^n} x = \frac{1}{n} \log_a x$ 1

(ii) Hence evaluate $\log_2 3 + \log_4 3 + \log_{16} 3 + \log_{256} 3 + \dots$ 2

QUESTION 7

(a) A particle P, of mass 2kg, is attached by a light inelastic string of length 1m to a fixed point A as shown in the diagram below. Another string of equal length attaches P to a smooth ring Q, of mass 3kg which is free to slide on a vertical wire that passes through A. The particle P is rotating in a horizontal circle of radius r , about the vertical wire with a constant angular velocity of 2π radians per second



Let T_1 represent the tension in the string PQ, T_2 the tension in the string AP and θ the angle of inclination of AP to the vertical wire.

(i) Copy the above diagram onto your paper and clearly indicate on your sketch all the forces acting on P and Q.

1

(ii) Write down the equations expressing the vertical and horizontal equilibrium of forces at points P and Q.

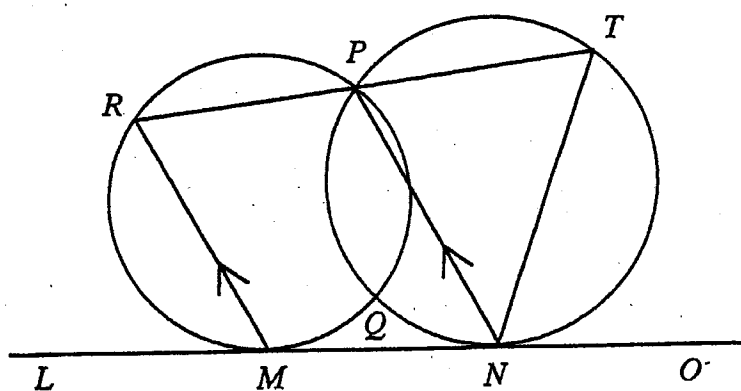
3

(iii) By using the equations in (ii) evaluate $\tan \theta$ in terms of r .

Hence calculate the vertical distance h of P below A ($g=9.8\text{ms}^{-2}$)

4

QUESTION 7 (cont)



(b) In the diagram the two circles intersect at P and Q. LMNO is a common tangent to the two circles. R is a point on one circle such that $MR \parallel NP$. RP produced meets the other circle at T.

(i) Copy the diagram.

(ii) Show that MNTR is a cyclic quadrilateral.

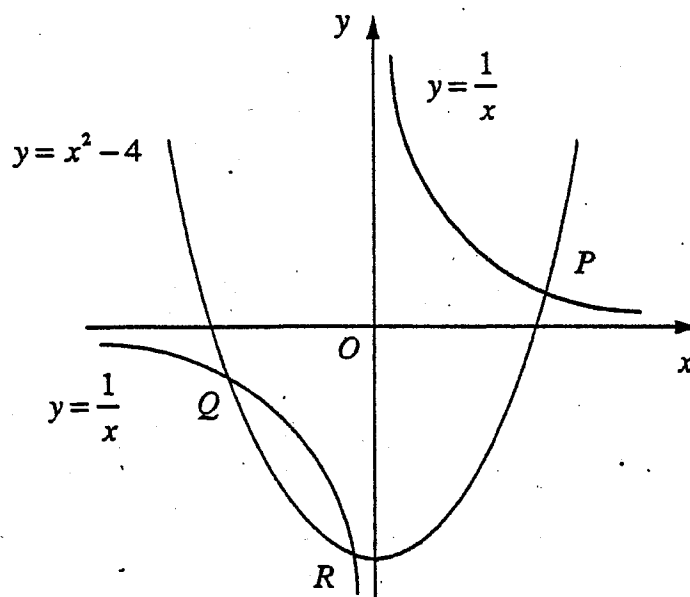
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(iii) G is the point of intersection of MT and NR. The circle through the points T, R, and G is drawn. Show that the tangent to this circle at G is parallel to MN.

3

QUESTION 8

(a)



The curves $y = x^2 - 4$ and $y = \frac{1}{x}$ intersect at the points P, Q, R where $x = \alpha, x = \beta, x = \gamma$

(i) Show that α, β, γ are the roots of the equation $x^3 - 4x - 1 = 0$

1

(ii) Find a polynomial equation with integer coefficients which has roots $\alpha^2, \beta^2, \gamma^2$.

2

(iii) Find a polynomial equation with integer coefficients which has roots $\frac{1}{\alpha^2}, \frac{1}{\beta^2}, \frac{1}{\gamma^2}$.

2

(iv) Hence find the numerical value of $OP^2 + OQ^2 + OR^2$.

2

(b) Newton's Method can be used to determine numerical approximations to the real roots of the equation $x^3 = 4$.

Let $x_1 = 2, x_2, x_3, \dots, x_n, \dots$ be a series of estimates obtained by iterative applications of Newton's method.

(i) Show that $x_{n+1} = \frac{2}{3}(x_n + \frac{2}{x_n^2})$.

2

(ii) Show algebraically that $x_{n+1} - \sqrt[3]{4} = \frac{(x_n - \sqrt[3]{4})^2(2x_n + \sqrt[3]{4})}{3x_n^2}$.

3

(iii) Given that $x_n > \sqrt[3]{4}$ show that $x_{n+1} - \sqrt[3]{4} < (x_n - \sqrt[3]{4})^2$.

2

(iv) Show that x_6 is accurate to 12 decimal places.

1

Question 1 Airraween HS Trial HSC

Extn 2

$$(a) \int \frac{dx}{\sqrt{(x+1)^2+4}}$$

$$= \ln(x+1+\sqrt{(x+1)^2+4}) + C$$

$$(b) 3x = a(x^2+2x+4) + (bx+c)(x+1)$$

$$x = -1$$

$$-3 = 3a$$

$$\therefore a = -1$$

$$x = 0$$

$$0 = 4a + c$$

$$\therefore c = 4$$

$$x = 1$$

$$3 = 7a + 2(b+c)$$

$$3 = -7 + 2b + 8$$

$$\therefore 2b = 2$$

$$b = 1$$

$$(ii) \int \frac{3x}{(x+1)(x^2+2x+4)} dx = \int \frac{-1}{x+1} + \frac{x+4}{x^2+2x+4} dx$$

$$= -\ln(x+1) + \int \frac{2x+2}{x^2+2x+4} + \frac{3}{x^2+2x+4} dx$$

$$= -\ln(x+1) + \frac{1}{2} \ln(x^2+2x+4) + 3 \int \frac{dx}{(x+1)^2+3}$$

$$= -\ln(x+1) + \frac{1}{2} \ln(x^2+2x+4) + \frac{3}{\sqrt{3}} \tan^{-1} \frac{x+1}{\sqrt{3}}$$

$$c) \int \tan^{-1} x \frac{d}{dx} (x) dx$$

$$= \left[x \tan^{-1} x \right] - \int \frac{x}{x^2+1} dx$$

$$(c) \frac{\pi}{4} - \frac{1}{2} \int_0^1 \frac{2x}{x^2+1} dx$$

$$= \frac{\pi}{4} - \left[\frac{1}{2} \ln(x^2+1) \right]_0^1$$

$$= \frac{\pi}{4} - \frac{1}{2} \ln 2$$

$$(d) \text{ let } x = \tan^2 \frac{t}{2}$$

$$dx = \frac{2}{1+t^2}$$

$$\text{when } x=0 \quad t=0$$

$$x=\frac{3}{4} \quad t=\sqrt{3}$$

$$\int_0^{\sqrt{3}} \frac{2}{5+t^2} dt$$

$$= \int_0^{\sqrt{3}} \frac{2}{5+5t^2+4-t^2} dt$$

$$= \int_0^{\sqrt{3}} \frac{2}{9+t^2}$$

$$= \left[\frac{2}{3} \tan^{-1} \frac{t}{3} \right]_0^{\sqrt{3}} = \frac{\pi}{9}$$

$$(e) \text{ let } u = \cos x \quad \frac{du}{dx} = -\sin x$$

$$\int (1-\cos^2 x) \cos^2 x \sin x dx$$

$$= \int (1-u^2) u^2 - du$$

$$= \int -u^2 + u^4 du$$

$$= -\frac{u^3}{3} + \frac{u^5}{5}$$

$$= -\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C$$

QUESTION 2

$$(a) (3-4i)(4+3i)$$

$$= 12+9i-16i+12$$

$$= 24-7i$$

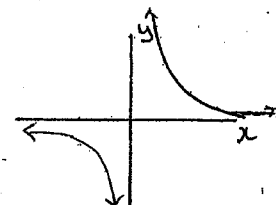
$$(b) \frac{1}{4+3i} \times \frac{4-3i}{4-3i} = \frac{4-3i}{25}$$

$$(c) (x+iy)^2 - (x-iy)^2 = 4i$$

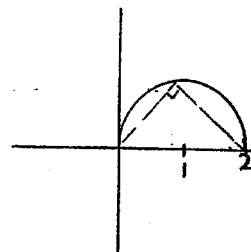
$$x^2+2xyi+y^2 - (x^2-2xyi+y^2) = 4i$$

$$4xyi = 4i$$

$$\therefore xy = 1$$



$$(ii) \arg(z-2) - \arg z = \pi/2$$



$$(c) i \bmod 2 = \sqrt{4}$$

$$= 2$$

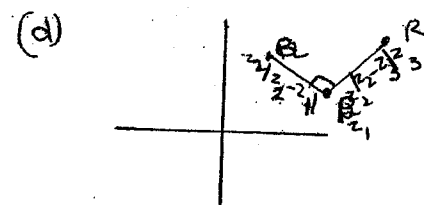
$$\arg z = \tan^{-1} \frac{1}{\sqrt{3}}$$

$$(u) 2^{-5} (\cos \pi/6 + i \sin \pi/6)^{-5}$$

$$= \frac{1}{32} (\cos -\frac{5\pi}{6} + i \sin -\frac{5\pi}{6})$$

$$= \frac{1}{32} (-\frac{\sqrt{3}}{2} - \frac{i}{2})$$

$$= -\frac{\sqrt{3}}{64} - \frac{i}{64}$$



ΔPQR is a right angled isosceles Δ $PQ = PR$

$$|z_2 - z_1| = |i(w_3 - w_1)|$$

$$= |w_3 - w_1|$$

multiplication by i rotates 90° anti-clockwise but does not change the length.

(e) Roots $1-2i, 1+2i$ third sum of roots $1-2i+1+2i+\alpha=3$

$$\therefore \alpha = 1$$

$$(x-1+2i)(x-1-2i)(x-1)$$

$$a^2 = 25 \quad b^2 = 16$$

$$16 = 25(1 - e^2)$$

$$e^2 = 1 - \frac{16}{25}$$

$$e = \frac{3}{5}$$

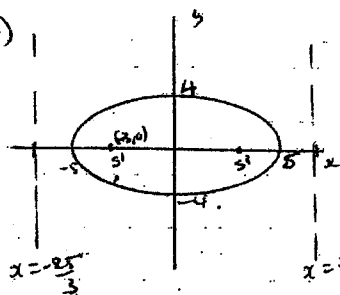
(1)

$$(u) \quad S(3,0) \quad S'(-3,0)$$

$$x = \pm \frac{25}{3}$$

(2)

(ii)



(1)

(iv) By definition

$$PS = e \cdot PM \quad \text{and} \quad PS' = e \cdot PM'$$

$$\begin{aligned} SP + S'P &= e(PM + PM') \\ &= e(M'M) \\ &= e \cdot \frac{2a}{e} \\ &= 2a \end{aligned}$$

which is independent of P
or in this case

$$SP + S'P = 10.$$

$$(b) \quad \frac{dx}{d\theta} = a \sec^2 \theta$$

$$\frac{dy}{d\theta} = b \sec^2 \theta$$

$$\frac{dy}{dx} = \frac{b \sec^2 \theta}{a \sec^2 \theta}$$

$$= \frac{b \sec \theta}{a \tan \theta}$$

$$y = \frac{b \sin \theta}{a \tan \theta} (x - a \sec \theta)$$

$$a \tan \theta y - ab \tan^2 \theta = b \sec \theta x - ab \sec^2 \theta$$

$$-a \tan \theta y + b x \sec \theta = ab(\sec^2 \theta - \tan^2 \theta)$$

$$\sec^2 \theta - \tan^2 \theta = 1$$

(2)

$$b x \sec \theta - a y \tan \theta = ab$$

$$(u) \quad \frac{dy}{dx} = -\frac{a \tan \theta}{b \sec \theta}$$

$$y - b \tan \theta = -\frac{a \tan \theta}{b \sec \theta} (x - a \sec \theta)$$

$$b \sec \theta y - b^2 \tan \theta \sec \theta = -a \tan \theta + a^2 \frac{\tan \theta}{\sec \theta}$$

$$\therefore \tan \theta \sec \theta$$

$$\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2 \quad (2)$$

(ii) when $x = 0$

$$A(0, -\frac{b}{\tan \theta}) \text{ tangent}$$

$$B(0, \frac{(a^2 + b^2) \tan \theta}{b}) \text{ normal}$$

(iv) foci of pair hyperbola

$$S(ae, 0)$$

If AB diameter A'S
is a right angle

$$m_{AS} = \frac{a \tan \theta}{a \sec \theta}$$

$$m_{BS} = \frac{(a^2 + b^2) \tan \theta}{(a^2 + b^2) \sec \theta}$$

$$m_{AS} \times m_{BS} = \frac{a \tan \theta}{a \sec \theta} \times \frac{(a^2 + b^2) \tan \theta}{(a^2 + b^2) \sec \theta}$$

$$= \frac{(a^2 + b^2) \tan^2 \theta}{a^2 \sec^2 \theta}$$

$$= \frac{(a^2 + b^2) \sin^2 \theta}{a^2}$$

$$e^2 = 1 = \frac{b^2}{a^2}$$

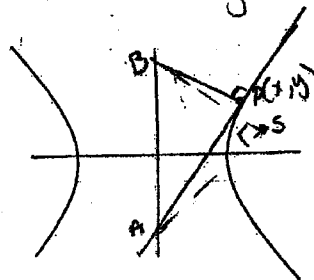
$$e^2 = \frac{b^2}{a^2} + 1$$

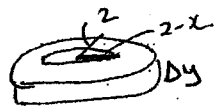
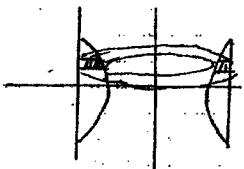
$$= \frac{b^2 + a^2}{a^2} \quad (3)$$

$$m_{AS} \times m_{BS} = -\frac{a^2}{a^2 + b^2} \times \frac{a^2 + b^2}{a^2}$$

$$= -1$$

So AB is diameter of circle
passes through foci.





Now $y^2 = 4 - x^2$
 $4x = 4 - y^2$
 $x = \frac{4 - y^2}{4}$

$$dV = [\pi 2^2 - \pi (2-x)^2] dy$$

$$V = \lim_{dy \rightarrow 0} \sum_{y=-2}^2 4\pi - \pi(2 - \frac{4-y^2}{4})^2 dy$$

$$= \int_{-2}^2 4\pi - \pi(2 - 1 + \frac{y^2}{4})^2 dy$$

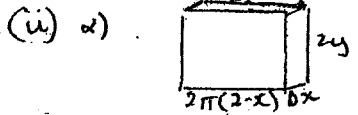
$$= \pi \int_{-2}^2 4 - (1 + \frac{y^2}{2} + \frac{y^4}{16}) dy \quad (b)$$

$$= 2\pi \int_0^2 3 - \frac{y^2}{2} - \frac{y^4}{16} dy$$

$$= 2\pi [3y - \frac{y^3}{6} - \frac{y^5}{80}]_0^2$$

$$= 2\pi [6 - \frac{8}{3} + \frac{32}{5}]$$

$$= \frac{89}{15} \pi$$



$$\Delta y = 2y \times 2\pi(2-x) \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^1 4\pi y(2-x) \Delta x$$

$$V = \int_0^1 4\pi \sqrt{4-x^2} (2-x) dx$$

$$= 8\pi \int_0^1 \sqrt{1-x} (2-x) dx \quad (2)$$

b) $u = 1-x \quad \frac{du}{dx} = -1$

$x=0 \quad u=1$ when $x=1 \quad u=0$

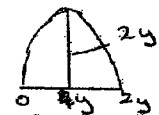
$$8\pi \int_1^0 (u+1) \sqrt{u} - du$$

$$= 8\pi \int_0^1 u^{3/2} + u^{1/2} du$$

$$= 8\pi [\frac{2}{5} u^{5/2} + \frac{2}{3} u^{3/2}]$$

$$= 8\pi [\frac{2}{5} + \frac{2}{3}] \quad (3)$$

$$= \frac{89}{15} \pi$$



(c) Simpson's rule is derived by fitting a parabolic arc to a set of 3 points. Hence it will give exact area for a parabola. (1)

(ii) $A = \frac{y}{3} [f(0) + 4f(y) + f(2y)]$

$$= \frac{y}{3} [0 + 8y + 0]$$

$$= \frac{8y^2}{3} \quad (2)$$

(iii) $dV = \frac{8y^2}{3} \Delta x$

$x^2 + y^2 = 100$
 $y^2 = 100 - x^2$

$$\int_{-10}^{10} \frac{8(100-x^2)}{3} dx$$

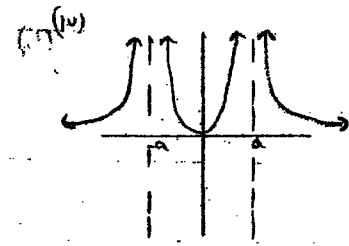
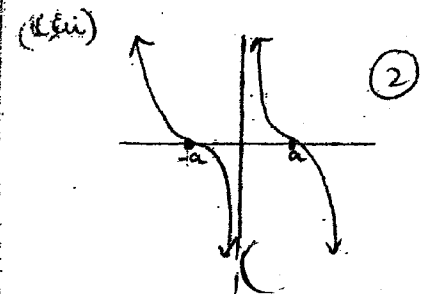
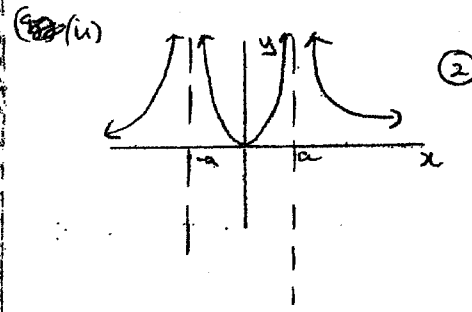
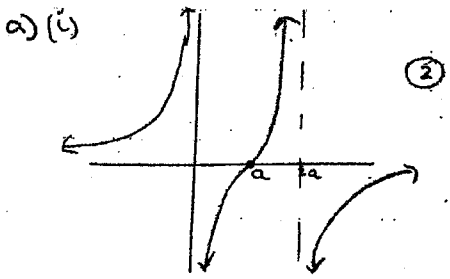
$$= \frac{16}{3} \int_0^{10} 8(100-x^2) dx$$

$$= \frac{16}{3} [100x - \frac{x^3}{3}]_0^{10}$$

$$= \frac{16}{3} [100 \times 10 - \frac{1000}{3}] \quad (3)$$

$$= 3555 \frac{5}{9}$$

QUESTION 5



(b) when $x=-1$

$$-1 + 4 - a + b = 0$$

$$a - b = 3$$

$$f'(x) = 3x^2 + 8x + a$$

$$x=-1 \quad f'(-1) = 3 - 8 + a$$

$$= 0$$

$$\therefore a = 5 \quad b = 2$$

(c) $kx^3 + kx - 2x = 0$

$$2x = kx^3 + kx$$

$$2x = kx(x^2 + 1)$$

$$\frac{2x}{x^2 + 1} = kx$$

(d) Look at intersection

of $y=kx$ with $y = \frac{2x}{x^2 + 1}$

If k negative only goes through $(0,0)$

$$\frac{dy}{dx} = \frac{(1+x^2)2 - (2x)(2x)}{(1+x^2)^2}$$

when $x=0 \quad \frac{dy}{dx} = 2$

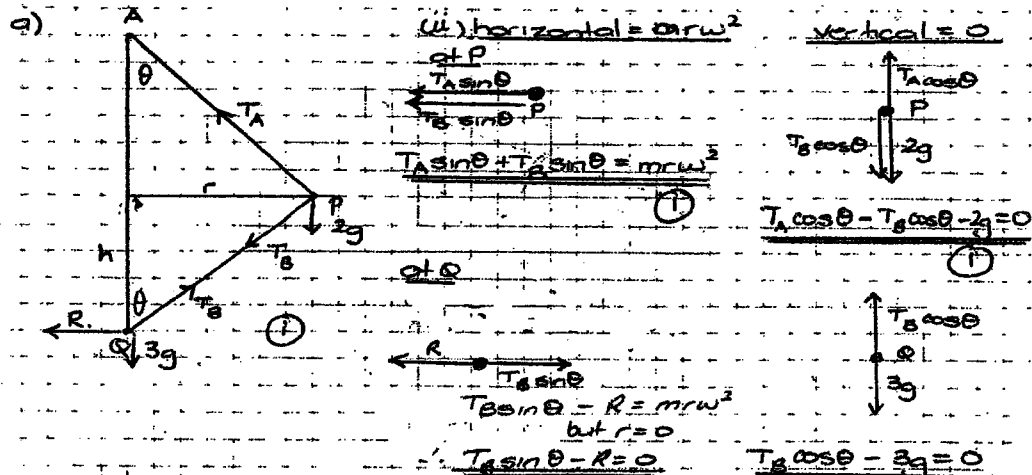
from $k=2$ to y -axis

only touches at one point

$$k > 2 \text{ and } k \leq 0$$

$$(3)$$

Question 7



$$(iii) \quad T_B \cos \theta = 3g$$

$$T_B = \frac{3g}{\cos \theta}$$

$$T_A \cos \theta - T_B \cos \theta = 2g$$

$$T_A \cos \theta - 3g = 2g$$

$$T_A \cos \theta = 5g$$

$$T_A = \frac{5g}{\cos \theta}$$

$$T_A \sin \theta + T_B \sin \theta = mr\omega^2$$

$$\frac{5g \sin \theta}{\cos \theta} + \frac{3g \sin \theta}{\cos \theta} = 2(r)(2\pi)^2$$

$$8g \tan \theta = 8\pi^2 r$$

$$\tan \theta = \frac{\pi^2 r}{g} \quad (3)$$

$$\frac{r}{h} = \tan \theta$$

$$= \frac{\pi^2 r}{g}$$

$$h = \frac{g}{\pi^2}$$

$$= 0.99 \quad (4)$$

as P moves it pulls on the ring, the ring thus exerts a horizontal force on the wire. By Newton's 3rd law the wire exerts an equivalent force on the ring, call it R .

QUESTION 8

$$(i) \quad \frac{1}{5x} = x^2 - 4$$

$$1 = x^3 - 4x$$

$$x^3 - 4x - 1 = 0 \quad (1)$$

$$(ii) \quad \text{let } y = x^2$$

$$x = \sqrt{y}$$

$$(\sqrt{y})^3 - 4\sqrt{y} - 1 = 0$$

$$y\sqrt{y} - 4\sqrt{y} = 1$$

$$\sqrt{y}(y - 4) = 1 \quad (2)$$

$$y(y - 4)^2 = 1$$

$$y(y^2 - 8y + 16) = 1$$

$$y^3 - 8y^2 + 16y - 1 = 0$$

$$(iii) \quad \text{let } y = \frac{1}{x}$$

$$\left(\frac{1}{x}\right)^3 - 8\left(\frac{1}{x}\right)^2 + 16\left(\frac{1}{x}\right) - 1 = 0$$

$$1 - 8x + 16x^2 - x^3 = 0 \quad (2)$$

$$x^3 - 16x^2 + 8x - 1 = 0$$

$$(iv) \quad P\left(x, \frac{1}{x}\right) \quad Q\left(\beta, \frac{1}{\beta}\right) \quad R\left(\gamma, \frac{1}{\gamma}\right)$$

$$OP^2 + OQ^2 + OR^2$$

$$x^2 + \beta^2 + \gamma^2 + \frac{1}{x^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$$

$$= 8 + 16$$

$$= 24 \quad (2)$$

$$(b) (i) \quad f(x) = x^3 - 4$$

$$f'(x) = 3x^2$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$x_{n+1} = x_n - \frac{x_n^3 - 4}{3x_n^2} \quad (2)$$

$$= x_n - \frac{x_n^3 - 4}{3x_n^2}$$

$$= \frac{2}{3} \left[x_n + \frac{4}{x_n^2} \right]$$

$$(ii) \quad \frac{(x_n - \sqrt[3]{4})^2 (2x_n + \sqrt[3]{4})}{3x_n^2}$$

$$= \frac{[x_n^2 - 2\sqrt[3]{4}x_n + (\sqrt[3]{4})^2][2x_n + \sqrt[3]{4}]}{3x_n^2}$$

$$= \frac{2x_n^3 + \sqrt[3]{4}x_n^2 - 4\sqrt[3]{4}x_n^2 - 2(\sqrt[3]{4})^2x_n + (\sqrt[3]{4})^3}{3x_n^2}$$

$$= \frac{2x_n^3 - 3\sqrt[3]{4}x_n^2 + 4}{3x_n^2}$$

$$= \frac{2}{3} \left[x_n + \frac{4}{x_n^2} \right] - \sqrt[3]{4}$$

$$= x_{n+1} - \sqrt[3]{4} \quad (3)$$

(iii) need to prove

$$\frac{2x_n + \sqrt[3]{4}}{3x_n^2} < 1$$

since $x_n > \sqrt[3]{4}$

$$\frac{2x_n + \sqrt[3]{4}}{3x_n^2} < \frac{2x_n + x_n}{3x_n^2}$$

$$< \frac{1}{x_n}$$

since $x_n > \sqrt[3]{4}$

$$\therefore \frac{1}{x_n} < 1 \quad (2)$$

$$x_{n+1} - \sqrt{4} < (x_n - \sqrt{4})^2$$

$$\begin{aligned} \text{(iv)} \quad x_6 - \sqrt{4} &< (x_5 - \sqrt{4})^2 \\ &< (x_4 - \sqrt{4})^4 \\ &< (x_3 - \sqrt{4})^8 \\ &< (2 - \sqrt{4})^{32} \\ &< 4.98 \times 10^{-13} \end{aligned}$$

$\therefore$ accurate to 12 decimal places has had 12 zeroes.

(1)